

Adapting Level of Traffic Stress for England: A Network-Scale Cycling Infrastructure Assessment in the West Midlands

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Abstract

Declaration

I hereby declare that this dissertation is all my own original work and that all sources have been acknowledged. It is XX,XXX words in length.

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Acknowledgements

Contents

Abstract	i
Declaration	ii
Acknowledgements	iii
List of Acronyms and Abbreviations	viii
1 Introduction	1
2 Literature Review	3
2.1 Infrastructure Quality, Perceived Safety, and Cycling Participation	3
2.2 LTN 1/20: Design Standards for Cycling Infrastructure in England	4
2.3 Level of Traffic Stress: Origins, Framework, and Applications	6
2.4 Growing Fragmented Cycling Networks	7
2.5 Research Gap and Questions	8
3 Study Area and Data	10
3.1 Study Area: The West Midlands	10
3.2 Data Sources: Overview	11
4 Methodology	13
4.1 Cyclable Network Extraction and Infrastructure Classification	13
4.2 Link-level LTS classification	15
4.3 Junction LTS Classification	17
4.4 Stress-Based Network Connectivity Evaluation	17
4.5 Network Growth Simulation	17
5 Results	18
5.1 Cyclable Network and Infrastructure Profile	18
5.2 LTS Classification	19
5.3 Stress-Based Network Density	19
5.4 Stress-Based Network Fragmentation	19
5.5 Stress-Based Network Reach	19
5.6 Network Growth	19
6 Discussion	20
6.1 Evaluating "Cycling for All": Network Connectivity Against LTN 1/20 Objectives	20

6.2	Junction Stress and the Structural Severance of the Low-Stress Network	20
6.3	Network Growth and Implications for Investment Coordination	20
6.4	Assessing OS NGD Cycle Lane Data: A First Independent Comparison with OpenStreetMap	20
6.5	Limitations and Future Work	20
7	Conclusion	21
	Bibliography	22
A	Supplementary Material	30
B	Research Log	31

List of Figures

1	Structure of the LTN 1/20 evaluation framework, comprising the Cycling Level of Service (CLoS) and the Junction Assessment Tool (JAT), and their combined role in determining Active Travel England funding eligibility. Compiled by author from Department for Transport DfT (2020a).	5
2	Study Area: The West Midlands. Left: location within England. Right: population distribution across the seven constituent metropolitan boroughs (Census 2021, 1 dot = 10 people), with hatched areas indicating rural Output Areas (OAs).	10
3	Analytical framework. LTN 1/20 design criteria (left) supply the classification thresholds operationalised on OS NGD Transport data (right) through Sections 4.1–4.3. The classified network is then evaluated against LTN 1/20's "Cycling for All" principle through connectivity analysis (Section 4.4) and network growth simulation (Section 4.5).	13
4	Directed cyclable network of the West Midlands Combined Authority by facility type. Where a road segment carries different provision by direction, the facility with the greater degree of separation from motor traffic is shown. The network is dominated by mixed traffic (grey) and off-road paths (purple). Physically separated infrastructure (blue) is concentrated along radial corridors into Birmingham and Coventry.	19

List of Tables

1	Level of Traffic Stress classification and corresponding population groups.	6
2	Datasets used in this study.	12
3	Highway Dedication types and legal cyclability under UK highway law.	14
4	Facility type classification derived from LTN 1/20 design hierarchy, mapped to OS NGD data sources.	15
5	Link-level LTS classification matrix derived from LTN 1/20 Safety subcriteria. .	16

List of Acronyms and Abbreviations

AADT	Annual Average Daily Traffic
CLoS	Cycling Level of Service
JAT	Junction Assessment Tool
KDE	Kernel Density Estimation
LCC	Largest Connected Component
LSOA	Lower layer Super Output Area
LTN 1/20	Local Transport Note 1/20
LTS	Level of Traffic Stress
MSOA	Middle layer Super Output Area
OD	Origin–Destination
OS NGD	Ordnance Survey National Geographic Database
OSM	OpenStreetMap
TfWM	Transport for West Midlands
WMCA	West Midlands Combined Authority

1 Introduction

Cycling has been widely recognised as a sustainable mode of urban transport, offering co-benefits across public health, air quality, carbon reduction, and potential congestion alleviation (WHO 2022; IPCC 2023; EU 2024). Central to realising these benefits is the provision of high-quality, connected cycling infrastructure, which is consistently associated with higher cycling participation (Hull and O’Holleran 2014; Pistoll and Goodman 2014; Buehler and Dill 2016; Mölenberg et al. 2019). Reflecting this evidence, the UK Department for Transport published Local Transport Note 1/20, Cycle infrastructure design (LTN 1/20), in July 2020, establishing the most comprehensive national standards for cycling infrastructure design and assessment applicable to England and Northern Ireland to date (DfT 2020a). LTN 1/20’s evaluation tools, the Cycling Level of Service (CLoS) and the Junction Assessment Tool (JAT), give practitioners a rigorous means of appraising individual schemes, assessing road segments and junctions against detailed design criteria through structured manual audit (DfT 2020a). What they do not address is the network-level question: whether compliant segments cohere into a connected whole. Network-level cycling connectivity is a property of whole journeys rather than individual schemes: a single high-stress link or junction can render an otherwise compliant route unusable for the very cyclists the guidance seeks to serve. There is therefore a need for a data-driven, network-level methodology capable of evaluating cycling infrastructure quality across a city or region using standard open and mapping agency data.

The Level of Traffic Stress (LTS) framework offers such a network-level approach, classifying road segments and junctions by the population group willing to tolerate their traffic conditions: from LTS 1, conditions suitable for children and old people, through LTS 2, tolerable to most adults, to LTS 3 and 4, which only progressively smaller and more confident minorities will accept (Mekuria et al. 2012). Because each stress level maps onto a population group, the classification links infrastructure quality directly to mode shift potential: expanding the network usable at lower stress thresholds is what brings the broader population into cycling. LTS-based network analysis has since become the dominant approach in cycling connectivity and equity research (Louro et al. 2026). Its non-compensatory logic, under which a single high-stress attribute determines classification, aligns naturally with how traffic conditions deter potential cyclists. Yet most published LTS criteria were calibrated to North American road design conventions, and applications to British cities have relied on simplified adaptations of these criteria rather than national design guidance (Cervero et al. 2019; Jeong and Smith 2025). No LTS classification grounded in UK design standards has been developed systematically.

The data foundations for such a classification have only recently emerged. The Ordnance Survey National Geographic Database (OS NGD) Transport Theme provides authoritative network

geometry with attributes directly relevant to stress classification, including carriageway width, legal access designation, and probe-derived vehicle speeds. Its Cycle Lane feature collection, which achieved full national coverage in March 2026, records cycle lane width, direction of cycling travel and infrastructure type with facility descriptions aligned to LTN 1/20 terminology (OS 2026a). The simultaneous availability of a statutory design standard and a nationally consistent authoritative dataset creates the conditions under which cycling networks across England can be classified and evaluated against a single national benchmark.

This dissertation develops a directed Level of Traffic Stress classification derived from LTN 1/20 design criteria, operationalises it on OS NGD Transport data, and applies it to evaluate network-level cycling conditions in the West Midlands. The region presents a revealing setting for such an evaluation. Among English regions it combines the highest share of private motorised trips with a cycling mode share of approximately 0.9% (WMCA 2024), yet its transport authority has committed to delivering a 500-mile network of low-stress cycle routes (WMCA 2020). The distance between this ambition and current network conditions is precisely what a standards-based classification is designed to measure. The study, conducted in collaboration with Transport for West Midlands, makes contributions to methodology and data: the first LTS framework calibrated to English design standards systematically, incorporating directed junction assessment and roundabout-specific criteria; and the first independent assessment of the OS NGD Cycle Lane dataset. The classified network is then used to evaluate connectivity against LTN 1/20's own principle of "cycling for all", proceeding from infrastructure supply to the journeys that infrastructure enables: the extent and spatial distribution of low-stress provision, the degree to which low-stress links cohere into connected components rather than isolated fragments, the reach available to residents from where they live, and the proportion of everyday travel demand, for commuting and for access to schools, that the network can carry at each stress threshold. A network growth simulation then examines how the sequence in which high-stress links are upgraded shapes connectivity outcomes, comparing selection strategies that exploit network structure against uninformed investment. Together these analyses illustrate the range of strategic questions that a nationally consistent, standards-based classification can support.

2 Literature Review

This chapter reviews the literature underpinning a standards-based, network-level cycling assessment. Section 2.1 establishes why infrastructure provision shapes participation, locating the mechanism in perceived safety and uneven stress tolerance. Section 2.2 examines LTN 1/20 and the limits of its scheme-level instruments. Section 2.3 introduces the Level of Traffic Stress framework and its applications. Section 2.4 reviews network growth simulation. Section 2.5 identifies the resulting gaps and states the research questions.

2.1 Infrastructure Quality, Perceived Safety, and Cycling Participation

The widely reported positive association between infrastructure provision and cycling participation raises a prior question of why provision should exert this influence (Hull and O’Holleran 2014; Pistoll and Goodman 2014; Buehler and Dill 2016; Mölenberg et al. 2019). A substantial body of work answers this by locating the mechanism in safety, and more precisely in its subjective dimension: whereas objective safety concerns the recorded incidence of crashes and injuries, it is the perceived safety, or how safe cycling is felt to be, that most directly shapes the decision to cycle, particularly for prospective and occasional riders (Chaurand and Delhomme 2013; Sanders 2015; Manton et al. 2016). This distinction between objective and subjective safety is what the language of traffic stress was introduced to capture. Early work characterised the discomfort imposed on a cyclist by the speed, volume, and proximity of adjacent motor traffic as a graded property of the road environment rather than a binary of safe and unsafe (Sorton and Walsh 1994; Landis et al. 1997; Harkey et al. 1998). Framed this way, the stress a cyclist experiences is understood as a systematic response to the traffic conditions a facility exposes them to, so that it can be inferred from the road’s physical and operational characteristics rather than measured directly. The value of the concept is that it makes the deterrent effect of traffic legible in terms of design variables that infrastructure can be built to modify.

Critically, tolerance of this stress is not uniform across the population. Stated-preference studies consistently find that groups currently under-represented among cyclists, women in particular, express stronger preferences for separation from motor traffic than men, with weaker but converging evidence for older adults and for those cycling with children (Aldred et al. 2017). The latent demand that infrastructure investment seeks to convert is therefore concentrated precisely among those least willing to tolerate high-stress conditions. This population corresponds to the widely reported “interested but concerned” segment, individuals who would consider cycling if protected from stressful traffic environments and who represent the largest and most winnable constituency for mode shift (Geller 2009; Mekuria et al. 2012; Dill and McNeil 2013). Consequently, the relevant question is not only whether provision works for those already riding

in traffic, but whether it also extends to the stress-averse users at the other end of the spectrum, on whom further uptake depends.

The relationship between perceived safety and cycling participation cannot be understood solely from the characteristics of individual links. Route choice evidence consistently shows that cyclists are willing to accept substantial detours to avoid stressful links and crossings (Broach et al. 2012; Park and Akar 2019; Berghoefer and Vollrath 2023), indicating that routes are evaluated as integrated wholes rather than as collections of independent facilities. Where no acceptable alternative exists, a single hostile link or crossing becomes a bottleneck that can render an otherwise suitable route unacceptable (Mekuria et al. 2012; Furth et al. 2016; Furth et al. 2018). It follows that the infrastructure quality relevant to the stress-averse majority is a property of connected networks rather than of individual facilities, and that its assessment requires criteria referenced to the populations expected to use them, applied to whole journeys rather than isolated schemes.

2.2 LTN 1/20: Design Standards for Cycling Infrastructure in England

“A travel revolution in our streets, towns and communities will have made cycling a mass form of transit. Cycling and walking will be the natural first choice for many journeys with half of all journeys in towns and cities being cycled or walked by 2030” (DfT 2020b).

This ambitious goal, set out in the Gear Change white paper, marked an unprecedented elevation in UK active travel policy. To translate this political commitment into consistent design practice, the Department for Transport simultaneously published Local Transport Note 1/20 (LTN 1/20), the most comprehensive national guidance for cycling infrastructure design and assessment to date (DfT 2020a). To cater for the broadest range of cyclists and deliver on Gear Change’s ambition of mass cycling uptake, LTN 1/20 establishes as first principle that cycle infrastructure should be accessible to everyone “from 8 to 80 and beyond”, with inclusive design running through all aspects of implementation (DfT 2020a).

To assess whether infrastructure meets this standard, LTN 1/20 organises its evaluation framework around five core design principles derived from Dutch practice (CROW 2016) — Coherence, Directness, Safety, Comfort, and Attractiveness — and introduced two complementary assessment instruments. The Cycling Level of Service (CLoS) tool appraises link-level attributes across 25 subcriteria under these five principles, each scored 0-2 with designated critical failures that override the overall assessment; the Junction Assessment Tool (JAT) complements CLoS at junctions, with its three scoring tiers explicitly anchored to distinct user populations (Figure 1). With the establishment of Active Travel England (ATE) as the national inspectorate in 2022, these instruments became central to a formal compliance process: only schemes achieving a minimum CLoS score of 70%, with no critical failures and no red-scored turning movements under the JAT, are generally considered eligible for Active Travel Fund allocations (DfT 2020a). This mechanism has effectively transformed LTN 1/20 from advisory guidance into a binding

quality threshold for publicly funded cycling infrastructure.

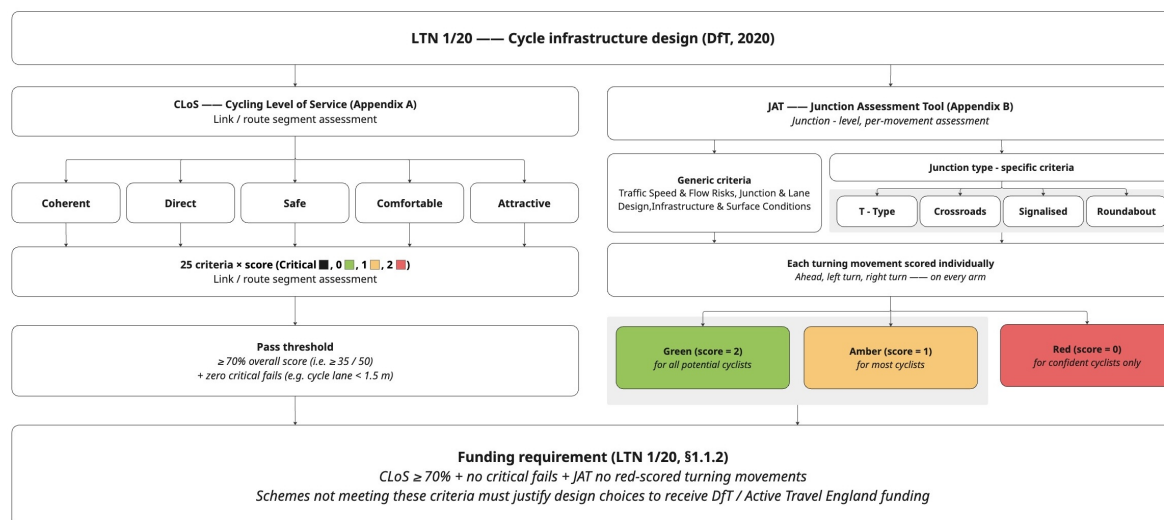


Figure 1. Structure of the LTN 1/20 evaluation framework, comprising the Cycling Level of Service (CLOS) and the Junction Assessment Tool (JAT), and their combined role in determining Active Travel England funding eligibility. Compiled by author from Department for Transport DfT (2020a).

Despite this comprehensive coverage, LTN 1/20's evaluation tools present three structural limitations when considered as a framework for network-wide cycling assessment. Firstly, the CLOS and JAT are scheme-level audit instruments requiring on-site evaluation by trained assessors, an approach that cannot scale to the classification of all cyclable links across an urban network and that risks inconsistency where subcriteria demand qualitative judgement (Hancock and Parkin 2024). Secondly, the CLOS abandons the population-referenced framing that the JAT adopts for its scoring tiers, defining all 25 subcriteria in purely technical terms. This reflects a structural constraint: the subcriteria mix stress-determinant factors that do implicitly delineate population acceptability, such as motor traffic speed, volume, and segregation, with experience-quality factors that do not, such as wayfinding and cycle parking provision. Additive scoring methods of this kind are inherently compensatory, such that strong performance on one set of criteria can arithmetically offset weaknesses in another (Pais et al. 2022; Wang et al. 2022); applied to CLOS, a link may therefore satisfy the 70% threshold through strong performance on the latter while falling short on stress conditions that, though not critical failures, would nonetheless deter the majority of potential cyclists. Thirdly, while the CLOS includes coherence subcriteria that address route-level continuity, these assess attributes of individual links rather than the connectivity of the network as a whole. They cannot answer whether a cyclist can complete a journey from origin to destination entirely within tolerable traffic conditions, yet end-to-end low-stress connectivity between origins and destinations has been identified as the most fundamental determinant of whether a cycling network can attract the widest possible segment of the population (Mekuria et al. 2012; Wang et al. 2016; Lowry and Loh 2017).

2.3 Level of Traffic Stress: Origins, Framework, and Applications

Where LTN 1/20 aspires to design for all from 8 to 80 but operationalises this through a broad composite of design criteria, the Level of Traffic Stress (LTS) framework (Mekuria et al. 2012) makes population acceptability the sole organising principle of its classification. Mekuria et al. operationalised this by adapting Geller's (2009) four-part population typology, the "strong and fearless," "enthused and confident," "interested but concerned," and "no way, no how," validated by Dill and McNeil (2013), into an infrastructure classification. They subdivided Geller's "interested but concerned" to distinguish children from adults, yielding four LTS levels each anchored to a distinct population group: LTS 1 corresponds to children, demanding the greatest separation from traffic; LTS 2 to the mainstream "interested but concerned" adult majority, with criteria anchored to Dutch facility standards; LTS 3 to the "enthused and confident," comfortable with moderate traffic interaction; and LTS 4 to conditions tolerable only by the "strong and fearless" (Table 1). The "no way, no how," who would not cycle regardless of provision, have no corresponding level. To classify segments, LTS draws on the small number of road characteristics most directly associated with cycling stress, notably motor traffic speed, facility type and width, and the nature of intersection approaches and crossings, using readily observable attributes such as the number of traffic lanes as a proxy for variables like traffic volume that are rarely available at network scale (Mekuria et al. 2012; Furth et al. 2016; Furth et al. 2018; Bearn et al. 2018). Classification follows a non-compensatory rule: a segment's LTS is determined by its worst-performing attribute, so that a single high-stress condition is sufficient to determine the classification regardless of other favourable characteristics.

Table 1. Level of Traffic Stress classification and corresponding population groups.

LTS Level	Population Group and Description
LTS 1	Suitable for almost all cyclists, including children cycling independently
LTS 2	Tolerable to the mainstream adult population; anchored to Dutch design standards proven to attract broad demographic participation
LTS 3	Tolerable to experienced cyclists who prefer dedicated space but accept moderate traffic interaction
LTS 4	Tolerable only to the strong and fearless riders willing to integrate with fast or heavy traffic

The original LTS framework proposed systematic network connectivity measures based on the population-anchored classification, quantifying the share of origin-destination pairs connected at each stress threshold and demonstrating how targeted gap closures can substantially multiply low-stress connectivity (Mekuria et al. 2012; Furth et al. 2016). This approach has since been widely adopted: a scoping review by Schön et al. (2024) identified 25 studies employing LTS to classify network links by stress level, primarily to measure cumulative accessibility by quantifying destinations reachable via low-stress links within defined distance or time thresholds (Murphy and Owen 2019; Wasserman et al. 2019; Gehrke et al. 2025). Beyond cumulative measures, LTS-based analysis has been used to evaluate job accessibility through detour-ratio methods that

permit travel on higher-stress links when low-stress routing is substantially longer (Furth et al. 2016; Lowry and Loh 2017; Cabral et al. 2019), to assess transport equity by linking low-stress connectivity to socioeconomic indicators (Tucker and Manaugh 2018; Kent and Karner 2019), and to identify network barriers and prioritise infrastructure investment (Lowry and Loh 2017; Putta and Furth 2019). Collectively, LTS modelling and network filtering have become the dominant approaches in cycling accessibility and equity research (Louro et al. 2026).

Two limitations constrain the applicability of existing LTS implementations to the UK context. Firstly, all published criteria were calibrated to North American road design conventions, speed regimes, and infrastructure types, and no LTS classification grounded in UK design standards has been developed systematically. Although the framework has been applied to British cities (Cervero et al. 2019; Jeong and Smith 2025), these implementations relied on simplified criteria derived from North American conventions rather than national design guidance. Secondly, the progressive simplification of the original criteria for network-scale application has produced considerable methodological divergence, with Harvey et al. (2024) finding substantial disagreement across seven LTS methods applied to the same cities, underscoring the sensitivity of results to variable selection and threshold definitions. These two limitations motivate the approach taken in this study: rather than adapting any single North American LTS variant, the classification is grounded in LTN 1/20, a single national design standard that provides both the threshold structure and the population-acceptability logic needed to anchor an LTS classification to English conditions.

2.4 Growing Fragmented Cycling Networks

Cycling networks in most cities remain sparse and fragmented (Schoner and Levinson 2014; Szell et al. 2022; Reggiani et al. 2023), so how they should expand, and in what order, is a live planning question. Network growth simulation addresses this by modelling expansion as the sequential addition of edges under a selection rule, generating investment trajectories whose connectivity outcomes can be compared (Barthélemy and Flammini 2009; Xie and Levinson 2009; Szell et al. 2022). Approaches to cycling network growth simulation can be divided by direction of construction: forward approaches iteratively add edges, either expanding from an existing network by bridging disconnected components (Natera Orozco et al. 2020), growing from scratch around topologically central seed points (Szell et al. 2022), or prioritising segments on the street network by potential routed cycling demand (Mahfouz et al. 2023). Inverse approaches begin from an idealised complete network and greedily remove the least important edges to generate budget-indexed subnetworks (Steinacker et al. 2022). Both directions can incorporate topological centrality (Szell et al. 2022) or cycling demand to weight edge importance (Steinacker et al. 2022; Mahfouz et al. 2023). A recurring finding is that how a fragmented network is grown matters as much as how much is built. Natera Orozco et al. (2020) show across fourteen cities that small, targeted investments connecting disconnected components rapidly consolidate a bicycle network, far outperforming random, uncoordinated addition, while Szell et al. (2022) find that uninformed

growth needs several times the investment of a flow-based strategy before a connected network emerges.

These studies operate on networks defined by infrastructure presence, in which an edge either belongs to the cycling network or does not, and they deploy growth simulation to identify an optimal expansion sequence. Neither feature carries over to a stress-classified network, where the object of investment is the upgrading of existing high-stress links, and low-stress conditions arise as much from quiet residential streets as from dedicated facilities.

2.5 Research Gap and Questions

The preceding review establishes three gaps at the intersection of design standards, network assessment, and investment planning. First, while LTN 1/20 provides the most comprehensive national design standard for cycling infrastructure in England, its evaluation instruments remain confined to scheme-level audit. The CLoS’s compensatory scoring obscures the population-referenced acceptability that the standard’s “cycling for all” principle implies, and neither the CLoS nor the JAT can assess whether compliant schemes cohere into a network that connects origins to destinations at tolerable stress levels (Section 2.2), despite end-to-end low-stress connectivity being the most fundamental determinant of a network’s capacity to attract the wider population (Mekuria et al. 2012; Lowry and Loh 2017).

Second, the LTS framework supplies precisely the network-level, population-anchored logic that LTN 1/20’s instruments lack, yet no published implementation is grounded in UK design standards. Existing applications to British cities rely on simplified adaptations of North American criteria (Cervero et al. 2019; Jeong and Smith 2025), and applied practice inherits further weaknesses from the Conveyal lineage: undirected max-propagation of junction stress and the absence of any treatment of roundabouts, a junction form both prevalent in the UK and associated with elevated cyclist risk (Section 2.3). The recent completion of the OS NGD Transport Cycle Lane collection removes the principal data barrier to a standards-grounded classification, but as a newly released dataset its completeness and reliability have not been independently assessed.

Third, the network growth literature demonstrates that the sequence of investment shapes connectivity outcomes as strongly as its volume, but existing simulations operate on networks defined by infrastructure presence, in which edges are added rather than upgraded (Section 2.4). Whether these findings transfer to a stress-classified network, where low-stress conditions arise as much from quiet streets as from dedicated facilities and the investment problem is one of reclassifying existing high-stress links, remains untested.

These gaps motivate the overarching aim of this dissertation: to develop and apply a network-level cycling infrastructure assessment grounded in English design standards. Three research questions structure the study:

- RQ1. How can the design criteria of LTN 1/20 be operationalised into a directed, network-scale Level of Traffic Stress classification, including junction and roundabout assessment, using OS NGD Transport data?
- RQ2. To what extent does the West Midlands cycling network deliver the connected, low-stress conditions that LTN 1/20's "cycling for all" principle requires, for the population groups the standard identifies?
- RQ3. How do alternative strategies for upgrading high-stress links compare in their capacity to improve low-stress network connectivity?

3 Study Area and Data

3.1 Study Area: The West Midlands

The study focuses on the West Midlands metropolitan region. Located in central England, it comprises the seven metropolitan boroughs of Birmingham, Coventry, Dudley, Sandwell, Solihull, Walsall, and Wolverhampton. With a population of over 3 million (ONS 2025), the West Midlands is the second biggest and second most populated conurbation in Britain after London (WMCA 2025). As Figure 2 illustrates, the region exhibits a polycentric urban structure, with Birmingham as the dominant core surrounded by smaller but densely populated centres including Wolverhampton, Coventry, and the towns of the Black Country¹. While the southern and eastern portions of Solihull and the fringes of Walsall and Dudley are predominantly rural. This mix of dense urban centres and lower-density peripheral areas presents distinct challenges for cycling infrastructure provision, as routes must negotiate fragmented land uses, varying traffic conditions, and considerable trip distances between centres (Delclòs-Alió and den Hoed 2024).

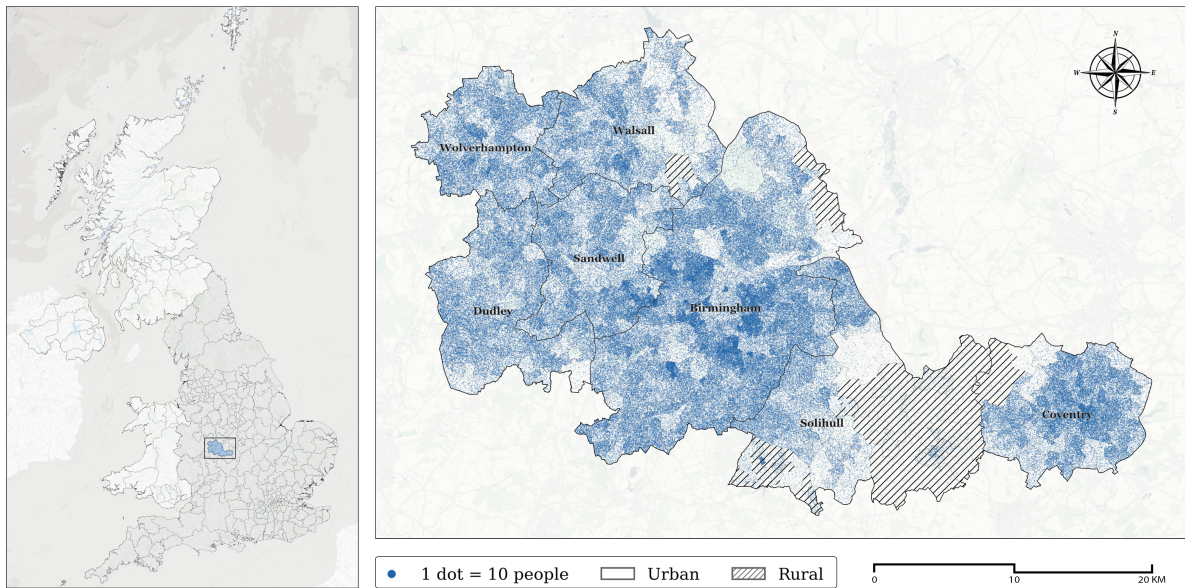


Figure 2. Study Area: The West Midlands. Left: location within England. Right: population distribution across the seven constituent metropolitan boroughs (Census 2021, 1 dot = 10 people), with hatched areas indicating rural Output Areas (OAs).

The West Midlands has been a centre of British automobile manufacturing since the early twentieth century, accounting for approximately 60% of national car production in the early 1970s (Donnelly et al. 2017). Its urban fabric bears this legacy: post-war interventions such as

¹The Black Country refers to the area immediately west and northwest of Birmingham, comprising the metropolitan boroughs of Dudley, Sandwell, Walsall, and Wolverhampton.

Birmingham’s Inner Ring Road, completed in 1971 as Britain’s first urban motorway (Gunn 2018), entrenched car-oriented mobility across the region, exemplifying what textcitemattioli2020 characterise as a self-reinforcing car-dependent system. National Travel Survey data show that the wider West Midlands region ² consistently records the highest share of private motorised trips among English regions, at 69% in 2024 (DfT 2025a). And cycling accounts for only approximately 0.9% of all trips in the metropolitan area (WMCA 2024), despite 41% of car journeys being under two miles (WMCA 2019).

In response, Transport for West Midlands (TfWM) has set out an ambitious active travel agenda centred on the Starley Network, a planned 500-mile network of traffic-free or physically separated cycle routes (WMCA 2020), aligned with the national Gear Change vision and LTN 1/20 design standards. Yet the gap between this policy ambition and the region’s measurable network-level cycling conditions remains poorly quantified, making the West Midlands a compelling case for the standards-based infrastructure assessment developed in this study.

3.2 Data Sources: Overview

The analytical pipeline draws on four principal data sources. The primary source is Ordnance Survey National Geographic Database Transport (OS NGD Transport), which provides network geometry, road classification, carriageway and cycle lane width, legal access designation, junction type and probe-vehicle-derived average speeds for constructing the directed, stress-classified network (Sections 4.1–4.3). The Cycle Lane feature collection, achieving full national coverage in March 2026 (OS 2026a), makes this study one of the first to incorporate authoritative national cycling infrastructure data into a network-level assessment. Department for Transport road traffic count data supply observed AADT volumes and training observations for minor road volume estimation (Section 4.2). OpenStreetMap (OSM) contributes cycling geometries for Path Link cross-validation, traffic signal locations for junction classification, and school locations for accessibility analysis (Sections 4.1, 4.3, 4.4). Census data from the Office for National Statistics provide MSOA-level commuting flows for OD connectivity and network growth evaluations, and LSOA-level population and boundaries for density and reach analyses (Sections 4.4–4.5). Table 2 summarises each dataset.

²The wider West Midlands region includes the seven metropolitan boroughs as well as the surrounding counties of Herefordshire, Shropshire, Staffordshire, Warwickshire, and Worcestershire.

Table 2. Datasets used in this study.

Dataset	Provider	Key Variables	Spatial Unit	Used In
Road Links	OS NGD	Geometry, classification, description, width	Edge	4.1–4.3
Path Links	OS NGD	Geometry, cycle lane presence	Edge	4.1
Highway Dedication	OS NGD	Legal access designation	Edge	4.1
Cycle Lane	OS NGD	Facility type, width, placement side	Edge	4.1–4.2
Average Speed (RAMI)	OS NGD	Probe-derived average speed	Edge	4.2
Road Node	OS NGD	Node form, geometry, description	Node	4.3
Path Node	OS NGD	Node geometry	Node	4.3
Road Traffic Counts	DfT	AADT by vehicle type	Count point	4.2
Cycling Network	OSM	Highway and cycleway tags	Way	4.1
Traffic Signals	OSM	Signal node locations	Node	4.3
Schools	OSM	Location	Point	4.4
Commuting Flows	ONS	OD pairs within 8 km	MSOA	4.4–4.5
Population	ONS	Resident population	LSOA	4.4
Boundaries	ONS	LSOA/MSOA polygons	Polygon	4.4–4.5

4 Methodology

This chapter presents the analytical framework for classifying and evaluating the West Midlands cycling network by Level of Traffic Stress (Figure 3). Sections 4.1–4.3 construct the directed, stress-classified network: extracting cyclable segments from OS NGD, assigning link-level LTS, and propagating junction stress with LTN 1/20’s criteria. Section 4.4 uses this network to test connectivity against LTN 1/20’s “Cycling for all” principle through density, fragmentation, and reach analyses. Section 4.5 compares three network growth strategies to inform upgrade prioritisation.

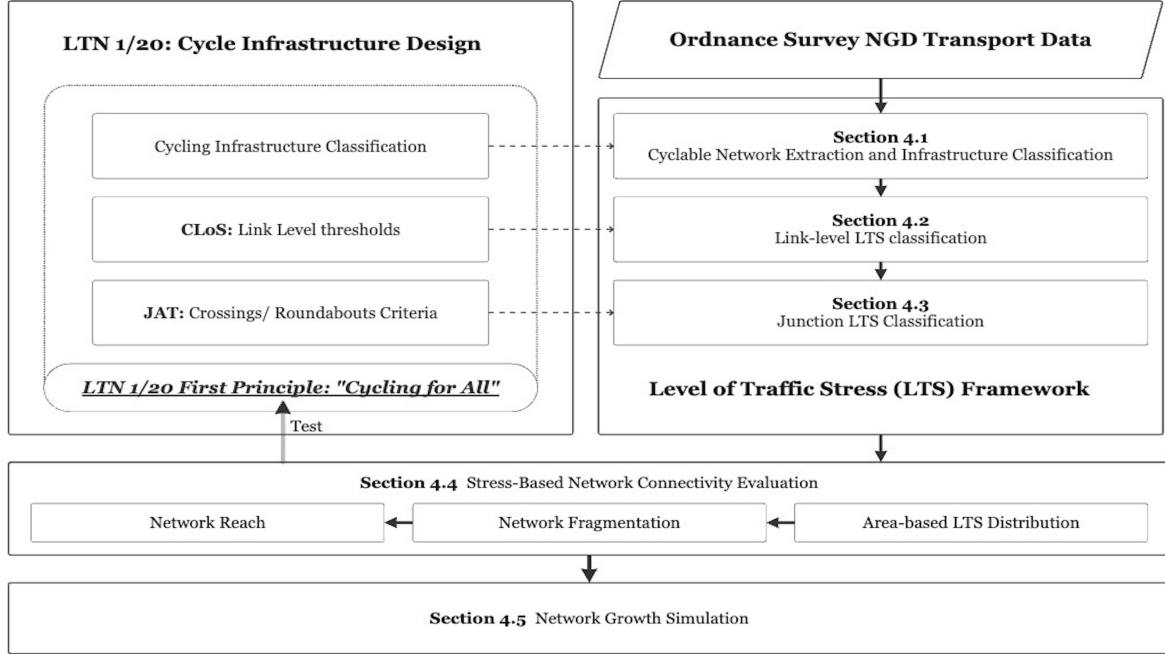


Figure 3. Analytical framework. LTN 1/20 design criteria (left) supply the classification thresholds operationalised on OS NGD Transport data (right) through Sections 4.1–4.3. The classified network is then evaluated against LTN 1/20’s “Cycling for All” principle through connectivity analysis (Section 4.4) and network growth simulation (Section 4.5).

4.1 Cyclable Network Extraction and Infrastructure Classification

The first stage of the classification process extracts all legally and practically cyclable segments from the West Midlands transport network. Cyclable infrastructure is represented in two feature collections under OS NGD Transport Theme: Road Links, which encode the carriageway network where cyclists may share space with motor vehicles (OS 2026f), and Path Links, which encode routes that are topologically independent from the carriageway, captured where they provide connectivity that cannot be inferred from the road network (OS 2026e). The legal cyclability of

segments is firstly checked through the Highway Dedication feature collection, which provides an indication of the type of highway user who has access to each section of the network, recording one of eight designation types (OS 2026d). Five of the seven designation types permit cycling under UK highway law, while ‘Motorway’ and ‘Pedestrian Way Or Footpath’ are excluded as non-cyclable (Table 3).

Table 3. Highway Dedication types and legal cyclability under UK highway law.

Highway Dedication	Cyclable?	Legal Basis
All Vehicles	Yes	Cycling permitted on public roads by default
Cycle Track or Cycle Way	Yes	Cycle Tracks Act 1984
Bridleway	Yes	Countryside Act 1968, s.30
Restricted Byway	Yes	Countryside and Rights of Way Act 2000
Byway Open to All Traffic	Yes	Wildlife and Countryside Act 1981, s.66(1)
Motorway	No	Motorways Traffic (England and Wales) Regulations 1982
Pedestrian Way or Footpath	No	No public right of way for cycling (cf. Highways Act 1835, s.72)

Note: The eighth type, ‘No Dedication Or Dedication Unknown’, is neither cyclable nor explicitly non-cyclable and is addressed through the description-based fallback classification for Road Links.

For Road Links, Highway Dedication covers most of the segments. Where a dedication record is absent or recorded as ‘No Dedication Or Dedication Unknown’, a fallback classification retains segments whose physical description feature indicates a carriageway type (‘Single Carriageway’, ‘Dual Carriageway’, ‘Roundabout’, ‘Slip Road’, ‘Shared Use Carriageway’, ‘Traffic Island Link At Junction’, and ‘Traffic Island Link’). Segments classified as ‘Motorway’ are excluded regardless of dedication status. This process yields 212,551 cyclable Road Links (97.8%), of which 84.2% are identified through Highway Dedication and 15.8% through the description fallback. For Path Links, Highway Dedication coverage is substantially lower, identifying only 10.1% of cyclable paths. A second layer retains paths where the feature data detects cycling infrastructure through its ‘presenceofcyclelane’ attribute, contributing a further 1.0%. No attribute in the OS NGD Path Link schema, however, provides a direct cyclability indicator. The Department for Transport’s transport connectivity metric reflects a comparable data source choice: while it constructs its driving network from Ordnance Survey data, it uses OpenStreetMap for cycling and walking networks, relying on OSM tags to determine which links can be walked or cycled on (DfT 2025b). Following this, a third layer performs spatial cross-validation against the OpenStreetMap cycling network, matching remaining paths without Highway Dedication or cycle lane presence to OSM cycling geometries within a 5-metre nearest-neighbour spatial join and recovering an additional 40,035 paths. Paths with explicitly non-cyclable Highway Dedications are excluded at all layers. This process yields 45,023 cyclable Path Links (40.3% of the total collection). Each segment is then expanded into directed edges, and cycle lane records are joined only to the edge whose travel direction aligns with the lane’s placement side, capturing the inherently asymmetric distribution of cycling infrastructure and avoiding overestimation of facility provision.

Each directed cyclable segment is then classified into one of eight facility types derived from LTN 1/20’s design hierarchy (Table 4). The OS NGD Cycle Lane dataset, whose type definitions

are aligned to LTN 1/20, records dedicated cycling infrastructure as separate features referencing their parent Road Link (OS 2026b; OS 2026c). A Road Link with a matching Cycle Lane record is classified according to its cycle lane description; where multiple records exist for the same link, the more protective facility governs. Road Links without a matching record, or where the cycle lane type is recorded as ‘Unknown Type Of Cycle Track Or Lane’, default to Mixed Traffic. All cyclable Path Links are classified as ‘Off-Road’, as they are topologically independent from the road network and carry no motor traffic speed or volume attributes (OS 2026e). Following classification, Road Links and Path Links are merged into a single unified and directed network for use in the subsequent LTS assessment.

Table 4. Facility type classification derived from LTN 1/20 design hierarchy, mapped to OS NGD data sources.

Facility Type	OS NGD Source	Classification Feature
Off-Road	Path Link	Cyclable Path Links
Shared Use	Cycle Lane	‘Unsegregated Shared Use Cycle Facility’ / ‘Segregated Shared Use Cycle Track’
Fully Kerbed Cycle Track	Cycle Lane	‘Fully Segregated Cycle Track’
Stepped Cycle Track	Cycle Lane	‘Stepped Cycle Track Along Road’
Light Segregation	Cycle Lane	‘Lightly Segregated Cycle Lane On Road’
Mandatory Cycle Lane	Cycle Lane	‘Mandatory Cycle Lane On Road’
Advisory Cycle Lane	Cycle Lane	‘Advisory Cycle Lane On Road’
Mixed Traffic	Road Link (main)	‘Unknown Type Of Cycle Track Or Lane’ or no matching Cycle Lane record

Note: Some facility types overlap conceptually. Off-Road paths may in practice function as shared-use routes or as kerbed cycle tracks. This does not affect the subsequent LTS classification, as all such facilities are fully separated from motor traffic.

4.2 Link-level LTS classification

To address the absence of an England-calibrated LTS classification identified in Section 2.3, the link-level LTS classification adapts the (Mekuria et al. 2012) framework by replacing its North American thresholds with those from LTN 1/20, the statutory design standard for cycling infrastructure in England and Northern Ireland. Although the Cycling Level of Service (CLoS) tool embedded in LTN 1/20 uses compensatory scoring across heterogeneous subcriteria, which prevents it from isolating population-level acceptability (see Section 2.2), the core Safety subcriteria addressing motor traffic speed (Criterion 10), volume (Criterion 11), segregation provision and carriageway width (Criterion 12) are themselves non-compensatory stress determinants constructed on CROW-derived logic (CROW 2016), directly consistent with the LTS framework. It is this subset that the present study selects for link-level LTS assignment. LTN 1/20 further provides a speed–volume–facility matrix that grades traffic conditions by the population they accommodate. This study maps that graduated logic onto the LTS framework: conditions suitable for all users including children correspond to LTS 1; those acceptable only to confident cyclists correspond to LTS 2; those suitable for few correspond to LTS 3; and conditions triggering a critical safety failure correspond to LTS 4 (Figure ??).

Table 5. Link-level LTS classification matrix derived from LTN 1/20 Safety subcriteria.

Facility type	LTS 1	LTS 2	LTS 3	LTS 4
Off-Road	All conditions	—	—	—
Shared Use	All conditions	—	—	—
Fully Kerbed Cycle Track	All conditions	—	—	—
Stepped cycle track	$V_{85} \leq 30$ mph	V_{85} 30–37 mph	≥ 37 mph	—
Light segregation	$V_{85} \leq 30$ mph	V_{85} 30–37 mph	≥ 37 mph	—
Mandatory cycle lane	Width ≥ 1.8 m, $V_{85} \leq 20$ mph, AADT $\leq 5,000$	Width ≥ 1.8 m, V_{85} 20–30 mph, AADT $\leq 5,000$	Width < 1.8 m; or V_{85} 30–37 mph; or AADT 5,000–10,000	$V_{85} \geq 37$ mph; or AADT $\geq 10,000$
Advisory cycle lane	Width ≥ 1.8 m, $V_{85} \leq 20$ mph, AADT $\leq 5,000$	Width ≥ 1.8 m, V_{85} 20–30 mph, AADT $\leq 5,000$	Width < 1.8 m; or V_{85} 30–37 mph; or AADT 5,000–10,000	$V_{85} \geq 37$ mph; or AADT $\geq 10,000$
Mixed traffic	$V_{85} \leq 20$ mph, AADT $\leq 2,500$	V_{85} 20–30 mph, AADT 2,500–5,000	V_{85} 30–37 mph; or AADT 5,000–10,000	$V_{85} \geq 37$ mph; or AADT $\geq 10,000$; or nearside lane 3.2–3.9 m and AADT $\geq 5,000$

Notes:

1. Where multiple criteria apply to a single segment, the overall LTS is determined by the worst-performing criterion (non-compensatory).
2. Thresholds are derived from the CLoS Safety subcriteria (Criteria 9–12, 15) in LTN 1/20 Appendix A, supplemented by the speed–volume–facility matrix in Figure 4.1, which the CLoS itself references as complementary guidance for determining appropriate separation levels.
3. Motor traffic volume is expressed as Annual Average Daily Traffic (AADT), used as a proxy for the passenger car unit (PCU) thresholds specified in LTN 1/20 Criterion 11. LTN 1/20 additionally specifies HGV composition thresholds ($> 5\%$ for critical fail, 2–5% for score 0); these could not be applied due to the absence of network-wide vehicle composition data and are acknowledged as a limitation (see section 6.5).
4. Rural exception: mixed traffic segments with $V_{85} \leq 30$ mph and AADT $\leq 1,000$ are classified as LTS 1, following LTN 1/20 Figure 4.1 Note 3.

The link-level LTS classification begins with facility type and proceeds through three primary variables: prevailing speed, road width, and motor traffic volume. OS NGD provides carriageway or cycle lane width at the link level through average and minimum width attributes. For cycle lanes, this study uses the separately recorded modal lane width rather than the minimum, as it better captures typical conditions along a segment. For mixed traffic on single carriageways, average road width is used to assess the critical nearside lane width range of 3.2 to 3.9 m associated with elevated close-pass risk.¹ Segments triggering this override are classified as LTS 4.

Prevailing speed is represented by the estimated 85th percentile speed (V_{85}), consistent with the measurement basis specified in LTN 1/20. V_{85} is preferred over posted speed limits because actual operating speeds frequently diverge from the limit depending on road geometry and

¹On one-way roads, total width serves directly as lane width; on two-way roads, lane width is approximated as half the total width, with the override applied only where AADT exceeds 5,000, as lower volumes allow drivers to overtake via the opposing lane.

enforcement, offering a more reliable measure of cycling stress (Fitzpatrick and Program 2003; European Commission 2021; DfT 2025c). V_{85} was estimated from the OS NGD average speed dataset (RAMI), which provides mean vehicle speeds derived from probe vehicle data across multiple time periods. The mean of observed average speeds across all available time periods was used as the representative link speed, and V_{85} was calculated by applying a 7/6 multiplier derived from CA 185 *Vehicle speed measurement* (National Highways 2019), which defines V_{85} as the mean speed plus one standard deviation and estimates the standard deviation as approximately one-sixth of the mean under the assumption of a normal speed distribution.

OS NGD does not include motor traffic volume attributes. To address this, major road volumes were obtained directly from DfT annual average daily traffic (AADT) count point data, while minor road volumes were estimated following the methodology of (Morley and Gulliver 2016). A Poisson GLM was constructed using road network characteristics extracted via pgRouting on the England OSM network as predictors and DfT count point observations as the dependent variable. Five-fold cross-validation ($n = 3,695$ count points) yields an R^2 of 0.604 ± 0.019 and Spearman's ρ of 0.803 ± 0.015 , considered sufficient for the coarse-grained LTS classification task. Predicted and observed AADT values were assigned to OS NGD links via nearest-network spatial join. For the small number of links where no match was achieved, volume was imputed using the median AADT of the corresponding road classification to ensure network-wide coverage.

4.3 Junction LTS Classification

4.4 Stress-Based Network Connectivity Evaluation

4.5 Network Growth Simulation

5 Results

This chapter reports the results of applying the framework from Chapter 4 to the West Midlands cycling network. Sections 5.1–5.2 profile the classified network by facility type and LTS level. Sections 5.3–5.5 evaluate connectivity against LTN 1/20’s “Cycling for all” principle through density, fragmentation, and reach analyses, and Section 5.6 compares three network growth strategies to inform upgrade prioritisation.

5.1 Cyclable Network and Infrastructure Profile

The directed cyclable network for the West Midlands comprises 23,301 km of edges across 480,228 directed links (bidirectional segments counted once per direction), derived from OS NGD Transport Network filtered by the cyclability criteria described in Section 4.1. Figure 4 maps the network coloured by facility type. Mixed Traffic dominates at 18,935 km (81.3%), followed by Off-Road paths at 4,014 km (17.2%), comprising segments with little or no motor vehicle traffic. Active investment in dedicated cycling facilities is minimal: the three physically separated types (fully kerbed cycle track, stepped cycle track, and light segregation) total just 49.7 km, accounting for 0.2% of the network. Paint-only provision (advisory and mandatory cycle lanes) adds a further 63.1 km (0.3%) but offers no physical protection from motor traffic (likely an underestimate, see Section 6.4).

Spatially, dedicated facilities are concentrated in Birmingham, where fragments of fully kerbed track are visible along radial corridors into the city centre. Coventry, the birthplace of the modern safety bicycle, shows the second-longest continuous stretches of separated infrastructure, a product of its Local Air Quality Action Plan which funded fully segregated cycleways to avoid introducing a Clean Air Zone charging scheme (Chaplin 2021; Coventry City Council 2023). Elsewhere the picture is stark: Sandwell, Dudley, Walsall, and Wolverhampton show very little physically separated provision, and Solihull’s network outside its town centre is entirely mixed traffic. Beyond segregated provision, off-road corridors such as greenways and residential streets are prevalent across the conurbation’s urban areas but thin markedly toward the rural fringe, while the low-stress network elsewhere depends largely on residential streets. This spatial inequality in low-stress infrastructure provision is a recurring theme in the connectivity analysis that follows in Section 5.5.

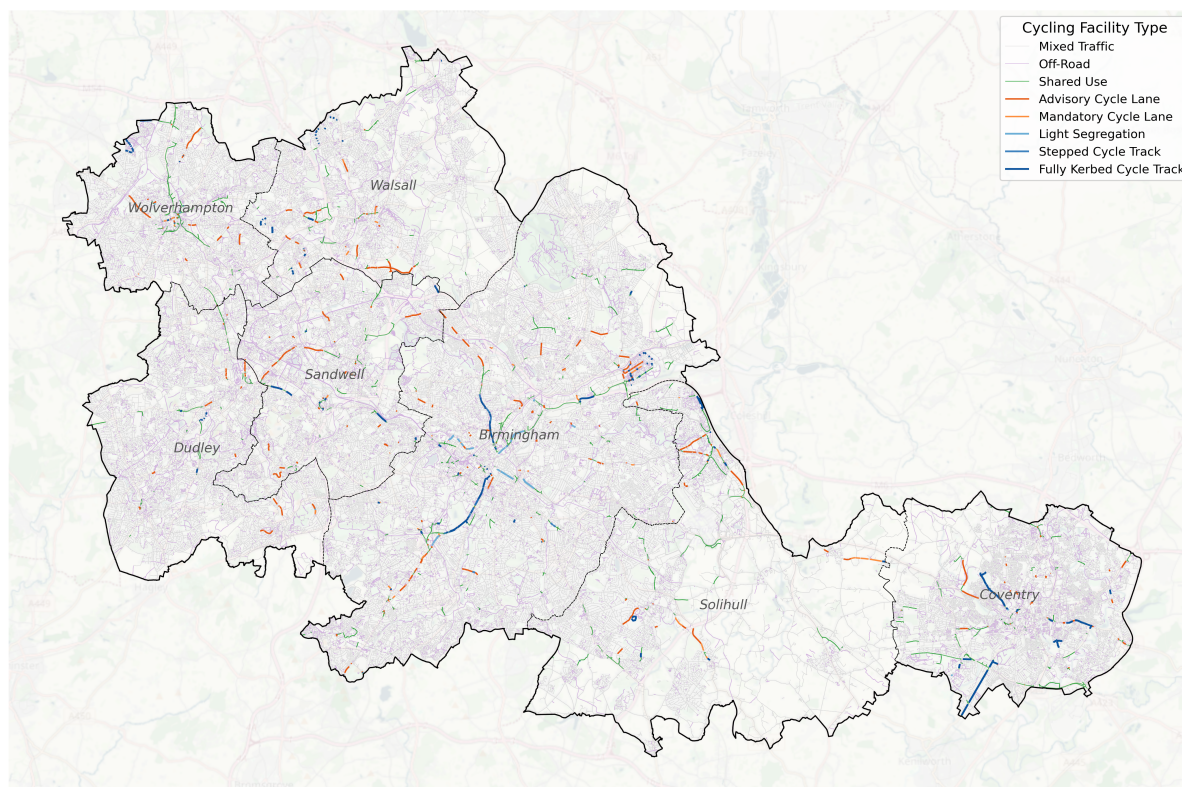


Figure 4. Directed cyclable network of the West Midlands Combined Authority by facility type. Where a road segment carries different provision by direction, the facility with the greater degree of separation from motor traffic is shown. The network is dominated by mixed traffic (grey) and off-road paths (purple). Physically separated infrastructure (blue) is concentrated along radial corridors into Birmingham and Coventry.

5.2 LTS Classification

5.3 Stress-Based Network Density

5.4 Stress-Based Network Fragmentation

5.5 Stress-Based Network Reach

5.6 Network Growth

6 Discussion

This chapter interprets the results in relation to the research aims and wider literature. Section 6.1 evaluates the West Midlands network against LTN 1/20's "cycling for all" principle, situating the connectivity findings within evidence on cycling demand and mode share. Section 6.2 examines junction stress as a structural mechanism of low-stress network severance, distinguishing the barrier roles of crossings and roundabouts. Section 6.3 draws implications from the growth simulation for investment prioritisation and cross-authority coordination. Section 6.4 assesses OS NGD cycle lane data through a first independent comparison with OpenStreetMap, and Section 6.5 discusses limitations and directions for future work.

6.1 Evaluating "Cycling for All": Network Connectivity Against LTN 1/20 Objectives

6.2 Junction Stress and the Structural Severance of the Low-Stress Network

6.3 Network Growth and Implications for Investment Coordination

6.4 Assessing OS NGD Cycle Lane Data: A First Independent Comparison with OpenStreetMap

6.5 Limitations and Future Work

7 Conclusion

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A Supplementary Material

B Research Log

Date	Type	Summary
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